

# OM SHAH

## Cybersecurity Graduate | Security Automation | Vulnerability Management

Sydney, NSW | [contact@omshahinfo.com](mailto:contact@omshahinfo.com) | [linkedin.com/in/omshah](https://linkedin.com/in/omshah) | [omshahinfo.com](https://omshahinfo.com) | [github.com/0m-5hah](https://github.com/0m-5hah)

### PROFESSIONAL SUMMARY

---

Cybersecurity graduate with three years of experience building Python automation in production and hands-on experience across external attack-surface monitoring, vulnerability assessment and security tooling. Reduced case handling time by 35% at Migrate Zone and built EightScope, a live external attack-surface monitoring platform with asset discovery, structured security findings and change tracking. Strong in Python, Nmap, Wireshark, web security and technical documentation.

### TECHNICAL SKILLS

---

**Security:** Vulnerability assessment, external attack-surface management, Nmap/NSE, Wireshark, DNS/TLS assessment, HTTP security headers, MITRE ATT&CK, OWASP Top 10, threat modelling

**Engineering:** Python, REST APIs, PostgreSQL, JSON/CSV, Excel automation, OCR workflows, Git/GitHub, HTML/CSS

**Platforms and tools:** Linux, Kali Linux, Windows, VMware, VirtualBox, AWS IAM and Security Groups, Burp Suite, Postman, Jira

### PROFESSIONAL EXPERIENCE

---

#### MIGRATE ZONE

December 2024 – Present

##### *Automation Engineer | Part-time*

- Reduced average immigration case handling time from approximately 8.0 hours to 5.3 hours, a 35% improvement, by co-leading modular Python automation for document validation, file naming, risk flagging and case routing.
- Consolidated financial audit checks into one workflow, removing duplicate processing and reducing manual cross-referencing across case files.
- Built an Excel prefill pipeline that mapped validated case information into lodgement templates, reducing repetitive entry and review work.
- Developed a lightweight monitoring tool that tracked government policy pages and surfaced relevant updates without repeated manual checks.

#### MIGRATE ZONE

July 2023 – November 2024

##### *Junior Automation Engineer | Part-time*

- Built Python scripts that automated case data entry and repetitive document-processing tasks across immigration workflows.
- Co-developed a PDF collation and OCR validation workflow that detected missing, outdated, lapsed and unreadable documents before lodgement, catching dozens of issues each month.
- Wrote internal process documentation that allowed non-technical employees to operate the automation toolkit without continuing developer support.

#### ITONKEY

August 2024 – October 2024

##### *Python Backend Developer Intern | Part-time*

- Built a Python system that converted structured business information into branded, multi-page PDF invoices using reusable templates.
- Contributed stock-tracking and basic reporting features to an inventory-management platform for small-business clients.

## SELECTED PRODUCT AND SECURITY PROJECTS

---

### EIGHTSCOPE

2026 – Present

Creator & Lead Developer | Independent Product · [eightscope.au](https://eightscope.au)

- Built and launched a live external attack-surface monitoring platform for Australian SMBs and managed service providers.
- Implemented DNS TXT ownership verification, subdomain discovery, hostname-level asset inventories and safety controls that block private, loopback, link-local and reserved destinations.
- Developed Quick, Standard and Full scan workflows covering DNS, email authentication, TLS, HTTP headers, exposed services, public files, technology detection and CVE signals.
- Designed structured findings with severity, confidence, evidence, remediation guidance and asset-aware scan comparison, with PostgreSQL-backed history and change tracking.

### SENTINEL

2025

Automated Network Vulnerability Scanner | UTS Capstone

- Built a Python vulnerability-assessment tool that used Nmap and NSE to discover hosts and services, map fingerprints to CPE identifiers and identify possible CVE matches using National Vulnerability Database data.
- Improved result quality by handling nested Nmap output, host-level script results, SMB and RDP fingerprints and common service-mapping edge cases.
- Generated structured, risk-prioritised JSON findings grouped by asset, including verification commands for further investigation.

### DNS TUNNELLING DETECTION LAB

2025

Independent Security Lab

- Built an isolated VMware lab to study DNS tunnelling behaviour and capture network evidence using Wireshark.
- Developed defensive detection logic using query entropy, subdomain length, burst timing and repeated-domain behaviour, mapped to MITRE ATT&CK T1048.003.
- Presented the captured PCAP through an interactive browser visualisation explaining how defenders could identify the traffic.

## EDUCATION

---

### BACHELOR OF CYBERSECURITY

2023 – 2026

University of Technology Sydney

Relevant study: System Security, Digital Forensics, Cloud Security, Cryptography, Cyber Threat Intelligence, Incident Response, Information Security Management and Network Security